

Education

The Chinese University of Hong Kong (CUHK)

PhD in Earth and Environmental Sciences

Advisor: Prof. Hongfeng Yang

Research focus: Earthquake rupture processes, finite-fault inversion, strong ground motion

Hong Kong, China

Sep 2026 – Present

Wuhan University, School of Earth and Space Sciences

B.S. in Geophysics

GPA: 3.87/4.00 (Rank: 2/23)

Wuhan, China

Sep 2022 – Jun 2026

Research Interests

- **Seismology:** Earthquake rupture processes, dynamic rupture modeling, seismic source inversion, finite-fault inversion, and induced seismicity.
- **Tectonics:** Fault systems, crustal deformation, plate boundary processes, and volcano-tectonic interactions.
- **Geodynamics:** Stress and strain evolution in the lithosphere, crust-mantle interaction, and deformation mechanisms.

Research Experience

Undergraduate Research

Jan 2023 – Jun 2026

Supervisor: Prof. Guoyan Jiang, Co-supervised by Postdoc. Yingwen Zhao, Wuhan University

- Utilized GAMMA software to process D-InSAR data (Sentinel-1) and extract coseismic surface deformation associated with the 2023 Xingwen Ms 4.9 earthquake sequence.
- Conducted inversion of fault geometry and slip distribution based on the derived surface deformation models.
- Analyzed earthquake catalogs to investigate spatiotemporal patterns of aftershocks.
- Researched correlations between aftershock activity and seismogenic fault.
- Applied the Epidemic-Type Aftershock Sequence (ETAS) model to assess earthquake triggering behavior within the sequence.
- Performed GNSS velocity field analysis to study regional crustal strain distribution.

Bachelor Thesis: Finite-Fault Inversion of the 2025 Myanmar Earthquake

2026

Supervisor: Prof. Guoyan Jiang, Wuhan University

- Conducted finite-fault inversion to resolve the rupture process and slip distribution of the 2025 Myanmar earthquake.
- Analyzed teleseismic waveform data to constrain fault geometry and rupture kinematics.

Publications

Mao GY, Zhao YW, Jiang GY, Dong HR, Zhao QS, Xu J and Li JL (2026). Seismogenic structure and rupture characteristics of the 2023 MS4.9 Xingwen Earthquake in Sichuan (China). *Earthquake Science*, 39. DOI: 10.1016/j.eqs.2026.03.001

Technical Skills

Programming: Python, MATLAB, C, C++, Shell

Tools and Software: GMT, SAC, ObsPy, Git, L^AT_EX, GAMMA

Languages: Chinese (native), English (fluent)